AfDB · DI4A · AFRICA AT NIGHT

Night-time lights

What darkness tells us — reading Africa's development from space

Concepts, instruments, artefacts and the approach of four hands-on labs: from raw radiance to a validated subnational indicator.



AFRICAN DEVELOPMENT BANK GROUP
GROUPE DE LA BANQUE AFRICAINE DE DÉVELOPPEMENT

Data Innovation for Africa

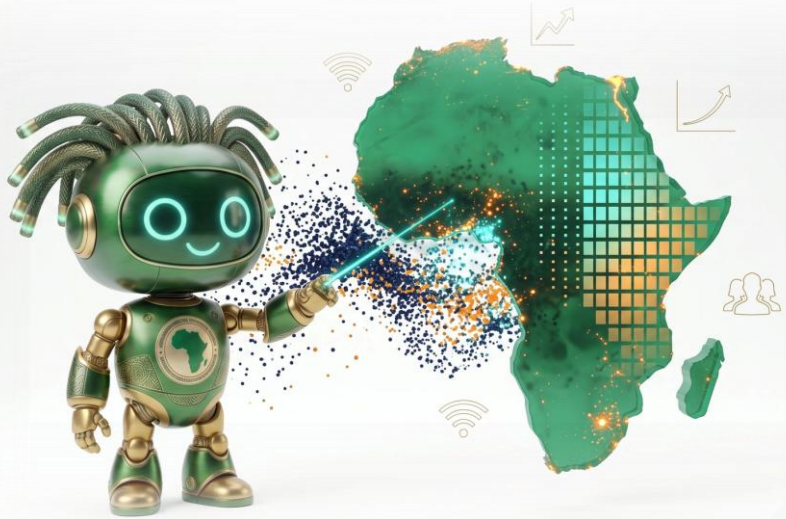
Explore DI4A online

SCAN TO VISIT THE WEBSITE

DI4A website

datainnov4africa-web.github.io/DI4A-Website

The DI4A programme, its Data Innovation Labs, use cases and resources — in one place.



SECTION 1

01

Foundations

Why would a photo of darkness teach us anything?

The core idea · Definitions · The chain of inference



Light betrays human activity

Almost everything we do after sunset needs light — and some of it escapes upwards, into a satellite's field of view.

This physical fact turns the night side of the planet into an observation of human presence that no census, survey or register matches on all four of these properties at once:

Consistent everywhere
Same method in every country

Continuous in space
Every cell of about 500 m

Politically neutral
Independent of what is reported

Freely accessible
Open NASA / EOG products

1992
Start of the continuous DMSP digital archive — three decades of history

500 m
VIIRS Day/Night Band pixel, nearly constant across the whole swath, data product

Every night
Global VIIRS overpass around 01:30 local time, since 2012

For an NSO: a census roughly every ten years, surveys every few years — lights arrive every night, for every district.

What NTL are — and are not

WORKING DEFINITION

Night-time lights (NTL) are measurements of the visible and near-infrared radiance emitted upwards from the Earth's surface, recorded by a satellite sensor during local night, in the absence of direct sunlight.

What they are

A physical measurement
Radiance in $nW \cdot cm^{-2} \cdot sr^{-1}$ for calibrated sensors (VIIRS).

A consistent grid
Same method in every country — in Rwanda as in Nigeria.

A near-real-time record
Daily since 2012; free monthly and annual composites.

What they are not

A photograph
"Earth at night" is a statistical composite of many cloud-free nights.

A direct measure of population or income
Those are inferences — each one must be justified.

A complete record of lighting
Indoor, shielded or sideways light does not reach the sensor.

From electricity to pixel: five links

1 — Energy and infrastructure

A grid, a generator or fuel exists and works.

2 — Activity after dark

Energy is used for lighting: streets, homes, markets, industry.

3 — Upward emission

Part of the light escapes to the sky instead of being shielded.

4 — Atmospheric transfer

Photons cross the atmosphere, attenuated by aerosols, vapour and clouds.

5 — Sensor response

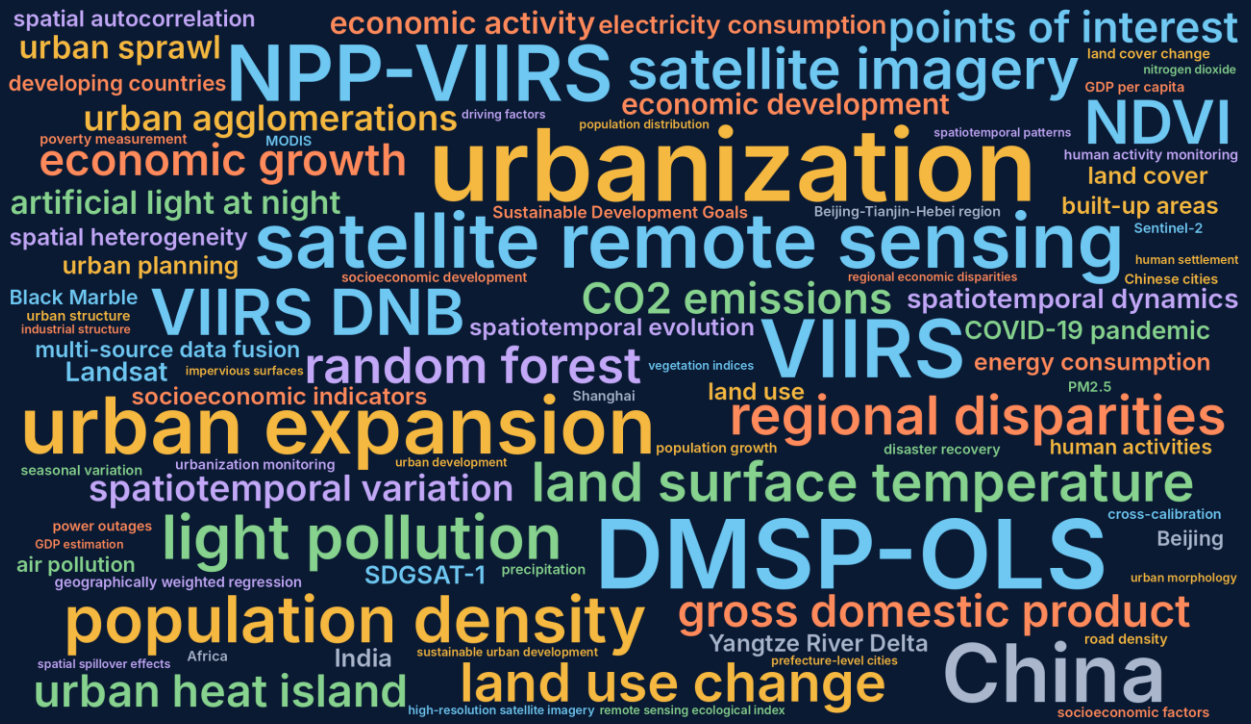
The detector integrates them over its band into a radiance value.

Where the chain breaks in an African context — a dark pixel can mean…

- **Link 1** — Load-shedding or fuel shortage — the people are there, the power is not
- **Links 2–3** — Lanterns and small solar kits too dim to be detected
- **Link 4** — Rainy-season clouds, harmattan, bush-fire smoke
- **Link 5** — Rural lighting below the sensor's detection threshold

Research keywords on night-time lights

Mots-clés de 4 450 publications OpenAlex sur les lumières nocturnes vues par satellite · plus le mot est grand, plus il est fréquent



4 450 publications trouvées (au 25 oct. 2026)

× 5 552 publications en 2025, contre 110 en 2016

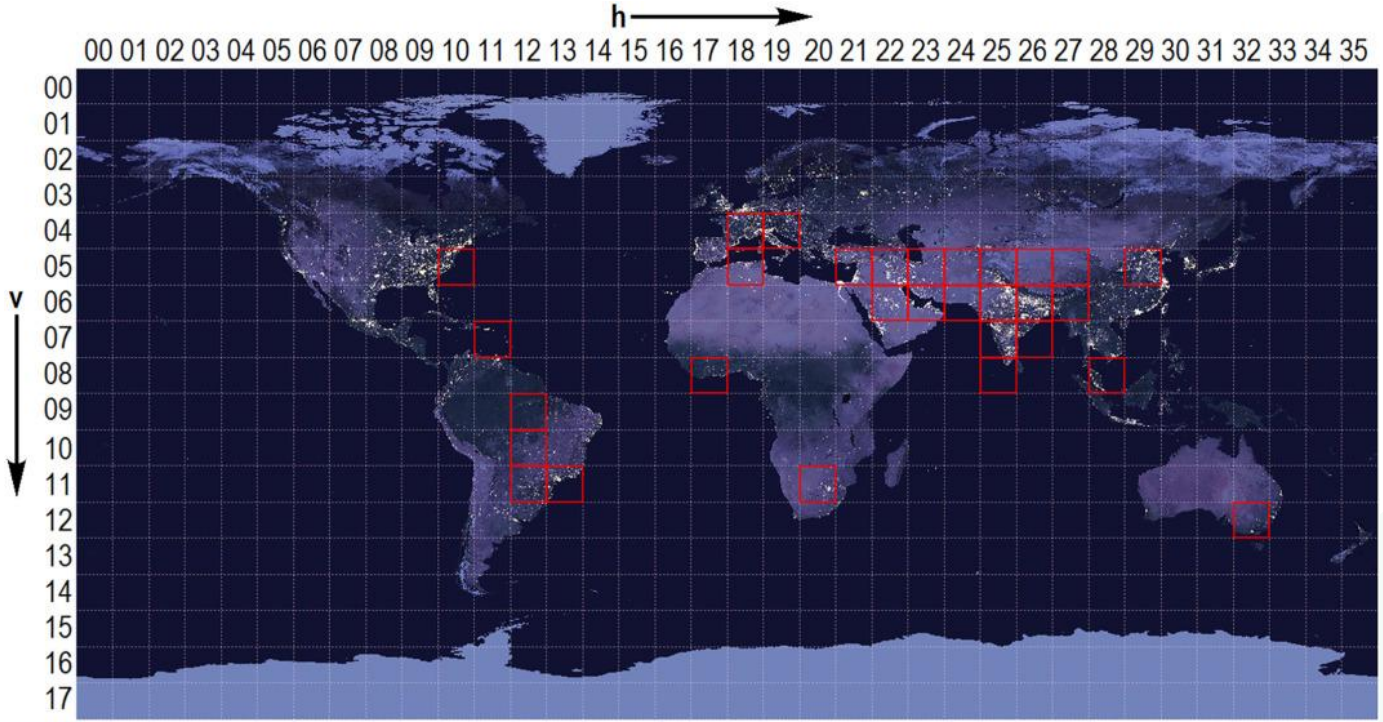
46 % avec un auteur affilié en Chine (17 % sans pays connu)

Thèmes (poids dans le nuage)

Capteurs et données	30,7 %
Urbanisation et territoire	26,8 %
Économie et développement	13,9 %
Environnement et chocs	11,9 %
Méthodes d'analyse	8,7 %
Lieux étudiés	8,0 %

Source : OpenAlex (api.openalex.org), extraction du 25/10/2026. Titre ou résumé contenant « nighttime / night-time light(s) » ou « night lights » et un terme satellitaire (VIIRS, DMSP, Black Marble, remote sensing…). Mots-clés en anglais tels qu'indexés ; terme de recherche retiré, variantes fusionnées.

A global grid of 10° × 10° tiles



SECTION 2

02

The instruments

What exactly is measuring, and at what resolution?

History · DMSP or VIIRS · Saturation and dynamic range



Fifty years of watching the dark

1970s — DMSP-OLS night imagery (film). City lights spotted while detecting clouds.

1992 — Start of the DMSP digital archive → annual "stable lights" series until 2013.

2011 — Launch of Suomi NPP with VIIRS: the first sensor calibrated for night-time radiance.

2012 — Cloud-free VIIRS monthly composites; start of the NASA Black Marble series.

2017 — Launch of NOAA-20 with a 2nd VIIRS; twin Black Marble products from 2018.

2018 — Luojia-1-01 demonstrates night imaging at ≈ 130 m from a small satellite.

2021 — SDGSAT-1, Glimmer imager: colour night imagery at 10–40 m.

2022 — Launch of NOAA-21: a 3rd VIIRS extends the series into the 2030s.

The irony: the first instrument was meant to see clouds by moonlight. City lights were noise — it took twenty years to turn them into the signal.

Two sensors, two different sciences

	DMSP-OLS · legacy	VIIRS Day/Night Band · modern
Period	1992 – 2013 (archive)	2012 – today (reference)
Radiometry	6-bit digital numbers (0–63), no on-board calibration	14-bit, calibrated radiance in nW·cm^{-2}·sr^{-1}
Resolution	≈ 2.7 km, 30 arc-second grid (≈ 1 km)	500 m, nearly constant across the swath
Saturation	City centres cap at 63	High dynamic range, three gain stages
Sensitivity	Misses faint rural lighting	Detects much fainter light
Products	Annual composites only	Daily, monthly, annual — open licence

New generation: Luojia-1-01 (≈ 130 m), SDGSAT-1 Glimmer (10 m panchromatic, 40 m colour) and commercial constellations distinguish streets and lamp types by colour.

The brightest, the least informative

Beyond a certain brightness, DMSP returns 63: suburb and city centre become identical.

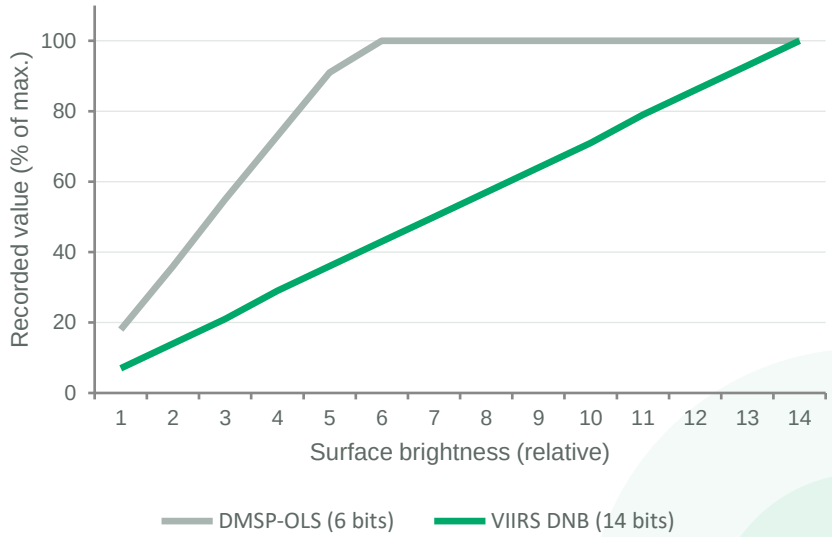
VIIRS spans about seven orders of magnitude, with three switchable gain stages.

Its low background noise makes a single lit fishing boat visible on a dark lake or ocean.

Read before any regression

Saturation is a ceiling, not noise that averages out. The information was never recorded — correcting it means modelling what the sensor did not see.

Sensor response to brightness (schematic)



Schematic illustration — data not fitted. VIIRS range after Elvidge et al. (2017).

SECTION 3

03

From photons to data

How does a night of observations become a publishable number?

Processing · Products · What shines · Indicators · Example



How a night-time image is made

1

Filter out clouds

Only cloud-free nights enter a composite. In the humid tropics, most can be discarded.

2

Remove the sun

Twilight, glint and instrument stray light are corrected or excluded.

3

Correct for the Moon

Moonlight reflected by the ground mimics emitted light; it is modelled and removed.

4

Drop the ephemeral

Fires, lightning, aurorae and boats are flagged by their instability.

5

Composite and mosaic

The retained nights are averaged into a monthly or annual grid, in tiles.

You will see these steps as layers in the Collect lab

VNP46A2 contains the corrected radiance and its evidence: cloud mask, lunar irradiance, snow flag and Mandatory_Quality_Flag (0 = high quality; 1–5 = poor, including aurorae and glint).

A product, not a snapshot

Two products for the same month can legitimately differ. Always record the product and its version.

Sources: Román et al. (2018); NASA LAADS DAAC, VNP46A2 v2 product page.

The NASA Black Marble product family

	VNP46A1 DAILY	VNP46A2 DAILY	VNP46A3 MONTHLY	VNP46A4 ANNUAL
Content	At-sensor radiance (top of atmosphere)	Moon-corrected NTL, gaps filled	Composite of corrected daily NTL	Composite of corrected daily NTL
Corrections	Uncorrected — inputs and geometry	Clouds, Moon, atmosphere, terrain, stray light	As A2 + number of observations and quality	As A2 + number of observations and quality
Typical use	Advanced users	Outages, reliability, events	Programme evaluation, seasonality	Long-term trends, subnational panels

Common to all: 15 arc-second grid (≈ 500 m at the equator) · 10°×10° tiles · HDF-EOS5 · since January 2012 · NOAA-20 twins named VJ146*.
Trap: radiance stored as integers — multiply by the 0.1 factor and mask the value 65535. **Alternatives:** EOG VNL composites; Google Earth Engine catalogue.

What really shines over Africa at night

Cities and towns
SIGNAL

Streets, buildings, markets, vehicles. The dominant and most studied component of the signal.

Corridors, ports and mines
SIGNAL

Roads, ports, airports, mining and industrial sites: the network between the bright spots.

Gas flares
MASK

Very bright isolated points in oil and gas areas — Niger Delta, Algeria, Libya. To be masked.

Light-attraction fishing
CONTEXT

Fishing with lights on Lakes Victoria and Tanganyika; fleets at sea.

Fires
MASK

Bush fires and agricultural burning: intense, brief and highly seasonal.

Natural or reflected light
REMOVE

Moonlight on bright sand or salt pans, airglow. No human signal.

Gas flares are catalogued separately by the VIIRS Nightfire product (Earth Observation Group).

Turning an image into a number

Fictional district · radiance per pixel

0.1	0.3	0.2	0.0
0.4	**2.5**	**6.0**	**0.6**
0.2	**4.0**	**12.0**	**1.5**
0.0	**0.8**	**1.2**	0.3

Coloured = above the 0.5 nW·cm⁻²·sr⁻¹ threshold

30.1

Sum of lights (SOL)

Total radiance within a boundary. Flagship indicator in the economics literature; the most sensitive to background noise.

8 px

Lit area

Pixels above a chosen threshold. Intuitive — but the threshold is the analyst's choice.

3.6

Mean radiance (lit)

Intensity independent of extent. Separates a district that brightens from one that spreads.

Threshold raised to 1.0 → lit area drops from 8 to 6 pixels (–25%). Always report product, version, boundary, threshold and masks — and test the threshold.

Illustrative values. Light concentration (radiance Gini) forms a 4th family, used for spatial inequality.

One question, end to end

Did the 2023 mini-grid programme really bring light to the targeted villages?

1 Translate the question into light

Electrified villages should have more nights above their background radiance after commissioning.

2 Choose the product

VNP46A3 monthly to date the change; VNP46A2 daily for reliability. Record the version.

3 Define the units

Settlement footprints plus a small buffer — not whole districts, where a few lit villages disappear.

4 Indicator and control

Mean radiance and share of lit nights; controls = similar villages outside the programme.

5 Read the difference

A break in programme villages, absent in the controls, is evidence; a common rise is regional.

6 Try to refute it

Check fires, flares, Moon and season; compare with the utility's connections or programme monitoring.

SECTION 4

04

Applications

Which statistical questions can these data really answer?

SDGs · Economy · Energy access · Urbanisation · Crises



Where lights support SDG monitoring

SDG 1

Poverty mapping

Small-area estimates combining lights and surveys

SDG 7

Electricity access and reliability

Locate unserved settlements; track lighting frequency

SDG 8

Economic activity

Subnational GDP breakdown; short-term activity signals

SDG 9

Infrastructure

Corridors, ports, industrial zones and special economic zones

SDG 11

Urbanisation

Urban footprint, sprawl or densification, new cities

SDG 16 · humanitarian

Conflicts and disasters

Outages, displacement signals, recovery monitoring

A complement, not a substitute. Lights bring spatial detail, frequency and independent verification to official statistics — they replace neither the census nor the survey.

Measuring the economy from orbit

What the research shows

Henderson, Storeygard and Weil (2012)

Lights carry real information about growth and can improve GDP where the accounts are weak.

Bruederle and Hodler (2018)

Lights reflect household wealth, education and health at the local level, based on DHS clusters in Africa.

Martinez (2022)

Lights serve as an independent benchmark to test the credibility of reported GDP growth.

Why the light–output link varies

- Sector structure: agriculture and mining create value with little light
- Income level: bright centres saturate or plateau
- Technology: the switch to LEDs changes light per watt
- Supply: load-shedding switches off lights, not output

WHERE LIGHTS ARE MOST USEFUL

Subnational | High frequency | Informal economy

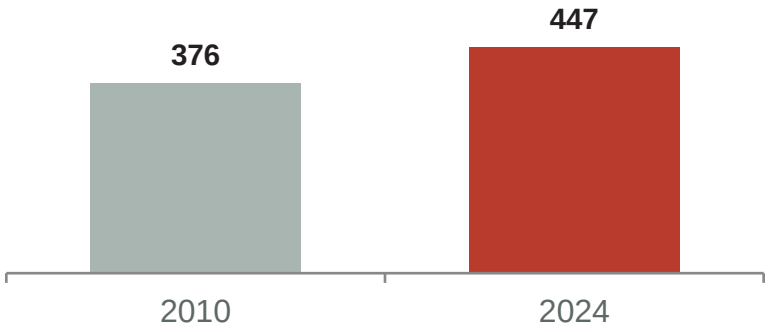
Sources: Henderson et al. (2012) AER 102(2); Bruederle and Hodler (2018) PLOS ONE 13(9); Martinez (2022) JPE 130(10).

Energy access and SDG 7: Africa's great challenge

655 M
people without electricity worldwide (2024)

> 560 M
of them in sub-Saharan Africa

Rural sub-Saharan Africa without electricity (millions)



Locate the unserved
Settlements visible by day but dark at night: communities probably not electrified.

Verify programmes
A grid extension that produces no light has, in practice, delivered nothing.

Measure reliability
Daily data show how often lights are on — the load-shedding that a "connected" status hides.

Plan the last mile
Distance to lit corridors guides the choice: grid, mini-grid or stand-alone solar.

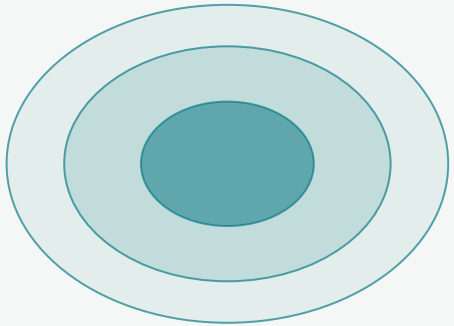
Mission 300 (World Bank Group and AfDB): 300 M people connected by 2030 — monitoring at that scale needs spatial evidence.

Sources: Tracking SDG 7 (IEA, IRENA, UN DESA, World Bank, WHO), 24 June 2026; HREA (Univ. of Michigan).

Urbanisation and the shape of cities

Sprawl

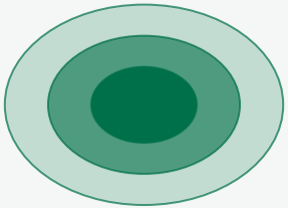
The footprint spreads outwards at low intensity



Area +40% · Radiance +6%

Densification

The same footprint intensifies



Area +6% · Radiance +40%

Same sum of lights — two very different cities (illustrative).

Map where registers are missing

Lights reveal the built-up footprint where cadastre and recent maps are lacking.

Tell sprawl from densification

Compare the growth of lit area with that of radiance — key for transport and services.

Spot new towns early

Mining towns, economic zones and displacement sites appear before the official geography does.

Pair with population grids

WorldPop or GHSL — a bright pixel is not a person.

Conflicts and disasters: the lights go out

Conflict · Sudan, 2023

After fighting broke out in Khartoum in April 2023, the capital's lights had fallen to noise level by May (EOG). UNOSAT mapped the outages in June 2023.

Conflict · Syria, 2011–2014

Li and Li (2014) measured a loss of more than 70% of night-time light nationally, tracking displacement province by province.

Disaster · Puerto Rico, 2017

After Hurricane Maria, daily Black Marble data mapped the outage and tracked recovery neighbourhood by neighbourhood.

Rule of interpretation: no light does not mean no people

Darkness can reflect displacement, destruction, fuel shortage, a curfew, a grid failure — or a decision to switch off street lighting. The satellite does not say which. Ground corroboration is essential and uncertainty must be reported.

Sources: Payne Institute / EOG (Sept. 2024); UNOSAT (June 2023); Li and Li (2014) IJRS 35(18); Román et al. (2019) on Puerto Rico.

SECTION 5

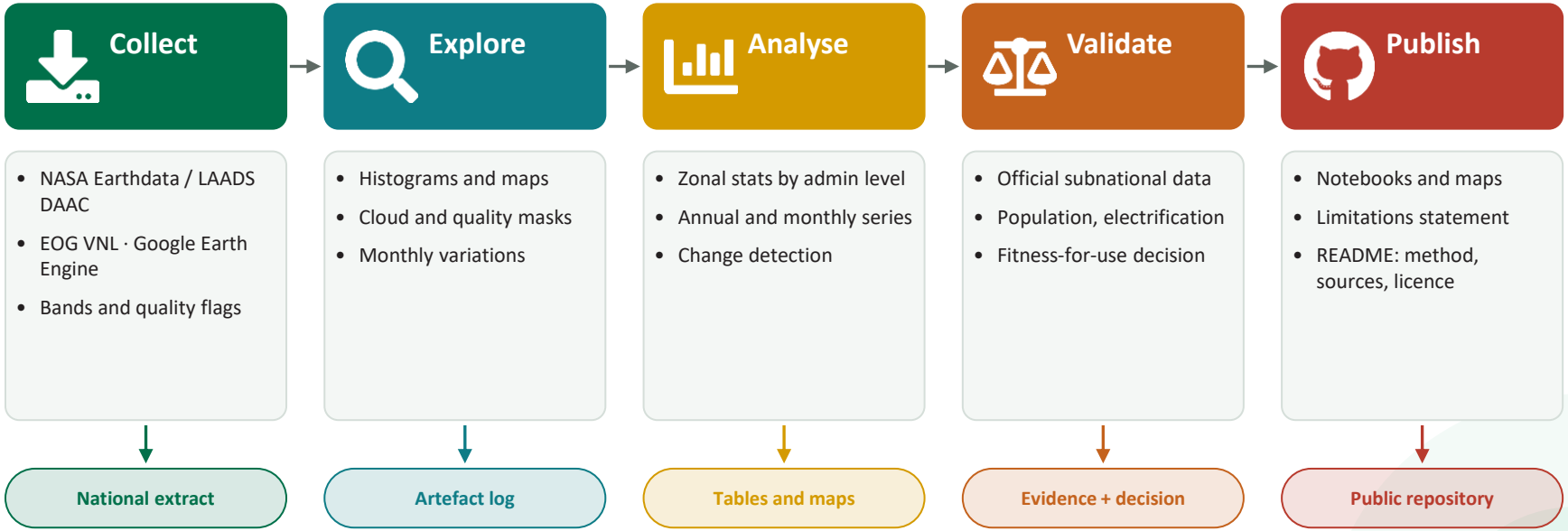
05

From lecture to lab



Approach · Choosing a product · Validation · Checklist · Discussion

Five steps, one reproducible pipeline



Example open-source tools: rasterio / xarray (reading), geopandas (boundaries), rasterstats or exactextract (zonal statistics), matplotlib (maps).

Which frequency for which question?

Annual

VNP46A4 · EOG VNL annual

STRENGTH

Cleanest series; noise and ephemeral lights largely filtered out.

WATCH OUT

Cannot date a change within the year.

TYPICAL NSO USE

Subnational panels, long-term trends, urban expansion, poverty maps.

Monthly

VNP46A3 · EOG VNL monthly

STRENGTH

Dates a change to within a season.

WATCH OUT

Rainy months may rest on very few valid nights.

TYPICAL NSO USE

Programme evaluation, seasonal activity, timelines of conflict and recovery.

Daily

VNP46A2

STRENGTH

Every night, corrected for Moon, atmosphere and terrain.

WATCH OUT

Large volumes; a single night is noisy — aggregate first.

TYPICAL NSO USE

Outages, disaster response, reliability of electricity supply.

Rule of thumb: start with annual. The finer the frequency, the more you must trust the corrections and manage noise.

Release, or keep as a diagnostic tool?

Q1 **Is the relationship strong and stable across units and years?**
Set the bar before you look.

Q2 **Have artefacts been identified and treated?**
Gas flares, blooming, seasonality, noise.

Q3 **Is the benchmark independent?**
WorldPop uses NTL as a covariate — circularity.

Yes to all three → Experimental statistic — release with its limitations

Partly → Diagnostic tool — internal use only

No → Not fit for purpose — document why

Limitations statement — template

Part of the deliverable, posted on GitHub

Data	Product, collection version, period, frequency
Choices	Boundaries, radiance threshold, masks applied
Artefacts	Which of the six were found, where, what treatment
Validation	Benchmark, year, number of units, strength of the relationship
Decision	Experimental, diagnostic or not fit — and why
Not to be used for	e.g. household income, population of small settlements

Indicative guide — adapt to your national quality assurance framework. Circularity: see Falchetta et al. (2019), Scientific Data.

Checklist for a sound analysis

1 — Name the product and version
VNP46A4 v2, VNL V2.x — they are not interchangeable.

2 — Don't splice naively
Any DMSP–VIIRS series needs a documented harmonisation.

3 — State and test the threshold
Show that the result holds over a reasonable range.

4 — Mask what you are not studying
Background noise, gas flares, light fishing, fires.

5 — Normalise on purpose
Per capita, per km², or not — say so and justify it.

6 — Validate on the ground
Census, surveys, utility data — independent sources.

7 — Prefer changes over levels
Differences cancel out much of the calibration error.

8 — Publish code and limitations
README with method, sources, licence and caveats.

THE GOLDEN RULE

State in one sentence what must be true about lighting behaviour for your conclusion to hold.

If you cannot write that sentence, you do not have a result yet.

Share: a documented, reusable repository is your institute's contribution to shared methodological guidance on new data sources.

IN CONCLUSION

Five ideas to take to the lab

1. Night-time lights are a proxy, never the quantity of interest — name the assumption that links them to your question.
2. The instrument defines the science: DMSP saturates, VIIRS is calibrated, and neither sees blue.
3. The data are worth most where official statistics are thinnest: subnational, high-frequency, crises, informal.
4. Darkness is not emptiness — off-grid lighting, clouds and load-shedding look alike from above.
5. Validate with independent sources, and publish your limitations with your results.



Darkness tells us a lot. It does not tell us everything — and it never speaks for itself.

The vocabulary of the labs

Radiance
Energy leaving a surface per unit area and solid angle; in $nW \cdot cm^{-2} \cdot sr^{-1}$ for NTL.

Digital number (DN)
Raw sensor count, with no physical unit; DMSP ranges from 0 to 63.

Composite
Grid assembled from many cloud-free nights — not a photo of one night.

Stable lights
Product from which fires, aurorae and other unstable sources have been removed.

Sum of lights (SOL)
Total radiance of all pixels within a boundary.

Blooming
Apparent spread of light beyond its source (scattering, geolocation).

BRDF correction
Adjustment for how reflected (lunar) light varies with viewing and illumination angles.

Zonal statistics
Summary of a raster's values (sum, mean, count) within each polygon.

Nadir
Point directly beneath the satellite, where the pixel footprint is smallest.

Skyglow
Diffuse brightening of the night sky by artificial light scattered in the atmosphere.

References and data sources

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DATA AND TOOLS

NASA Black Marble
LAADS DAAC · Earthdata

Earth Observation Group
VNL composites, Nightfire

Google Earth Engine
Hosted, analysis-ready archive

HREA
Electricity access by settlement

Open Nighttime Lights
World Bank tutorials